



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal to S.P. SAMY AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

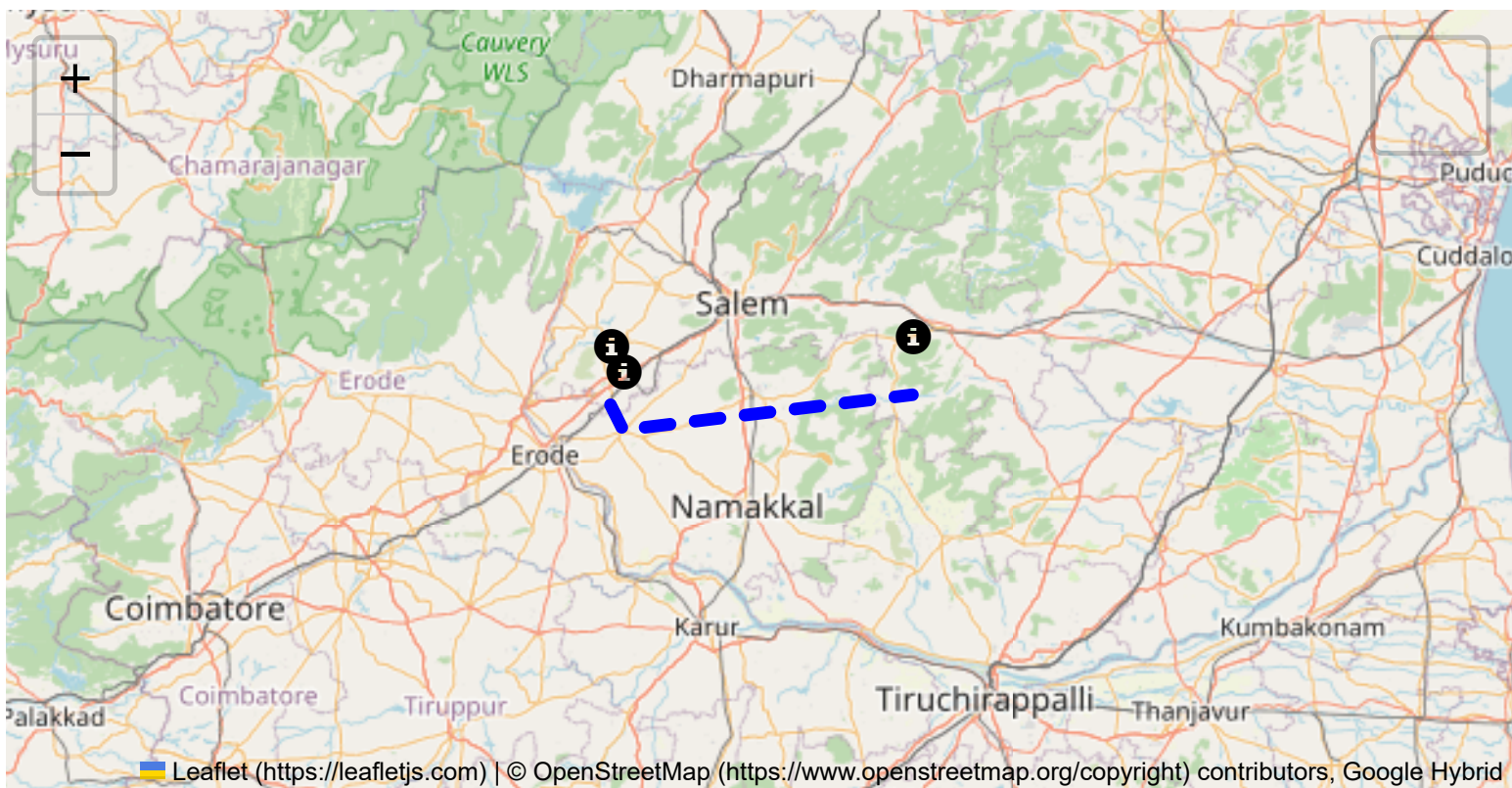
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

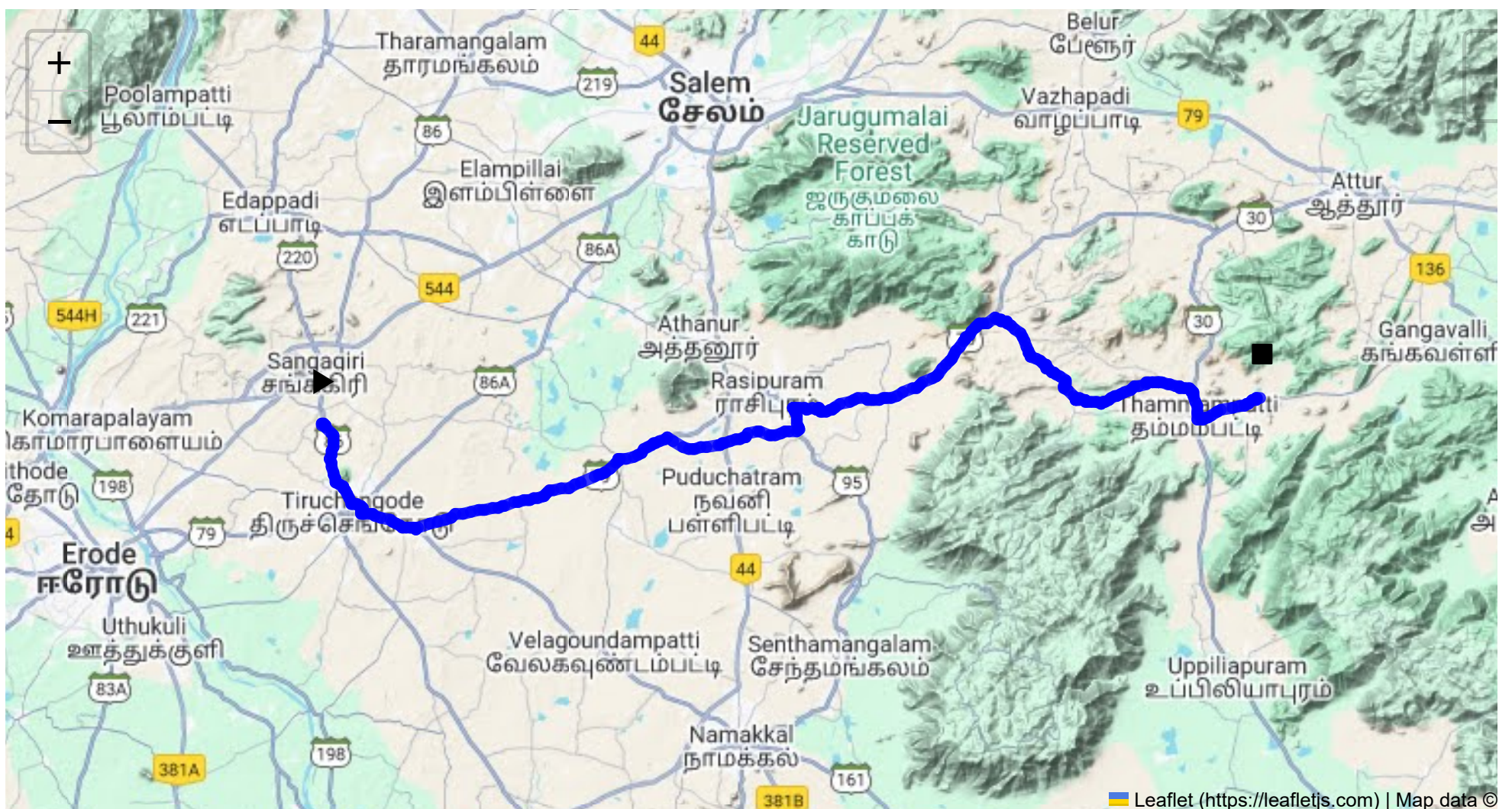
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 91.54 km**  
**Estimated Duration: 2.2 hours**  
**Adjusted Duration (Heavy Vehicle): 2.7 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.457178, 78.529823)**



## Welcome to the Journey Risk Management Study

### Route Safety Analysis Report

#### 1. Overview of the Route Map

The route starts at CVQF+23W in Sangagiri and proceeds to FG5J+432 in Salem via the Tiruchengode - Salem Main Road and the village of Thokkavadi. This journey spans approximately 91.54 kilometers. The



route primarily consists of state highways and rural roads, incorporating several small towns and agricultural areas.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences a tropical climate with a significant amount of rainfall from the northeast monsoon (October to December). During these months, heavy rainfall could lead to waterlogging and reduced visibility. Conversely, the summer months (March to June) often see extreme heat, potentially causing road surface damage and increasing vehicular overheating risks.

## 3. Analysis of Traffic Patterns

Traffic tends to be heavier during the morning (8:00-10:00 AM) and evening (5:00-7:00 PM) rush hours, especially around urban areas like Salem. Notable congestion points include the Tiruchengode market areas and intersections near Salem city limits. Heavy vehicle operators should anticipate slower travel during these periods.

## 4. Assessment of Road Quality and Infrastructure

The road quality varies, with better-maintained stretches on the state highways and less-developed sections through rural areas. Narrow lanes, potholes, and lack of proper signage are common concerns on secondary roads. Frequent road repairs can also cause temporary bottlenecks.

## 5. Suggestions for Alternative Routes for Emergencies

In case of congestion or roadblocks, an alternative route could involve taking State Highway 86 (SH-86) via Namakkal before connecting back to the main road toward Salem. This detour might be longer but can be useful if the primary route is compromised.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu enforces strict regulations on the transport of hazardous materials, including mandated escort vehicles for certain classes of dangerous goods. Compliance with these laws is mandatory, and trucks should be equipped with proper placarding and documentation.

## 7. Overview of Historical Incidents

The route generally has a low incidence of major accidents involving heavy vehicles or hazardous materials. However, minor incidents related to vehicular collisions due to inadequate road conditions have been reported, often influenced by adverse weather.

## 8. Environmental Considerations and Sensitive Areas

The route passes through agricultural zones and small settlements. Particular care should be taken to avoid contaminating water bodies and farmland in case of spills. No significant wildlife habitats or

ecological reserves are immediately affected by the route.

## 9. Analysis of Communication Coverage

Mobile network coverage is generally reliable along this route with some possible dead zones in rural areas. Coverage may vary between mobile network providers, so having multiple communication options is advisable.

## 10. Estimated Emergency Response Times

Emergency response times can vary, with faster responses possible within urban areas like Salem (approximately 30-45 minutes). In rural locales, especially during bad weather or peak traffic periods, response times could exceed one hour.

## 11. An Overall Summary of Risk Assessment

Overall, the route is moderately challenging, presenting typical hazards associated with regional roads in Tamil Nadu. Weather conditions and traffic can significantly impact travel time and safety, particularly for vehicles carrying hazardous materials. Proper planning, adherence to regulations, and situational awareness are crucial for mitigating risks.

The route analysis provides essential insights for training truck drivers to navigate this path efficiently while ensuring safety and adherence to regional guidelines.

### Risk Assessment - Turns

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 0  | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1  | Turn       | High       | 11.43968, 77.87345 | 15 KM/Hr    |
| 2  | Turn       | High       | 11.44019, 77.87547 | 15 KM/Hr    |
| 3  | Turn       | Medium     | 11.39685, 77.88861 | 30 KM/Hr    |
| 4  | Turn       | Medium     | 11.38346, 77.89975 | 30 KM/Hr    |
| 5  | Turn       | Medium     | 11.38302, 77.89995 | 30 KM/Hr    |
| 6  | Turn       | High       | 11.38188, 77.90257 | 15 KM/Hr    |
| 7  | Turn       | High       | 11.37830, 77.90160 | 15 KM/Hr    |
| 8  | Turn       | Medium     | 11.36754, 77.93919 | 30 KM/Hr    |
| 9  | Turn       | Medium     | 11.41361, 78.08073 | 30 KM/Hr    |
| 10 | Turn       | High       | 11.41453, 78.08062 | 15 KM/Hr    |
| 11 | Turn       | Medium     | 11.41476, 78.08094 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 12 | Turn       | High       | 11.42956, 78.11590 | 15 KM/Hr    |
| 13 | Turn       | Medium     | 11.42185, 78.13013 | 30 KM/Hr    |
| 14 | Turn       | Medium     | 11.42450, 78.15278 | 30 KM/Hr    |
| 15 | Turn       | Medium     | 11.43432, 78.20586 | 30 KM/Hr    |
| 16 | Turn       | High       | 11.43542, 78.20754 | 15 KM/Hr    |
| 17 | Turn       | Medium     | 11.45075, 78.20391 | 30 KM/Hr    |
| 18 | Turn       | High       | 11.45099, 78.20397 | 15 KM/Hr    |
| 19 | Turn       | High       | 11.44946, 78.22270 | 15 KM/Hr    |
| 20 | Turn       | High       | 11.44793, 78.22320 | 15 KM/Hr    |
| 21 | Turn       | Medium     | 11.44753, 78.22443 | 30 KM/Hr    |
| 22 | Turn       | Medium     | 11.45361, 78.23597 | 30 KM/Hr    |
| 23 | Turn       | Medium     | 11.49628, 78.32342 | 30 KM/Hr    |
| 24 | Turn       | Medium     | 11.50868, 78.33972 | 30 KM/Hr    |
| 25 | Turn       | High       | 11.51180, 78.34422 | 15 KM/Hr    |
| 26 | Turn       | Medium     | 11.51040, 78.34875 | 30 KM/Hr    |
| 27 | Turn       | Medium     | 11.50969, 78.34967 | 30 KM/Hr    |
| 28 | Turn       | Medium     | 11.50905, 78.35285 | 30 KM/Hr    |
| 29 | Turn       | Medium     | 11.48662, 78.36896 | 30 KM/Hr    |
| 30 | Turn       | Medium     | 11.47977, 78.38044 | 30 KM/Hr    |
| 31 | Turn       | Medium     | 11.47754, 78.38114 | 30 KM/Hr    |
| 32 | Turn       | Medium     | 11.46956, 78.39365 | 30 KM/Hr    |
| 33 | Turn       | Medium     | 11.46783, 78.39421 | 30 KM/Hr    |
| 34 | Turn       | High       | 11.46455, 78.39289 | 15 KM/Hr    |
| 35 | Turn       | High       | 11.45614, 78.40096 | 15 KM/Hr    |
| 36 | Turn       | Medium     | 11.45560, 78.40272 | 30 KM/Hr    |
| 37 | Turn       | Medium     | 11.45658, 78.40442 | 30 KM/Hr    |
| 38 | Turn       | Medium     | 11.45483, 78.40829 | 30 KM/Hr    |
| 39 | Turn       | Medium     | 11.46741, 78.45087 | 30 KM/Hr    |
| 40 | Turn       | High       | 11.46729, 78.45447 | 15 KM/Hr    |
| 41 | Turn       | Medium     | 11.46808, 78.45535 | 30 KM/Hr    |
| 42 | Blind Spot | Blind Spot | 11.46798, 78.45844 | 10 KM/Hr    |
| 43 | Blind Spot | Blind Spot | 11.46857, 78.45844 | 10 KM/Hr    |
| 44 | Turn       | High       | 11.46799, 78.45992 | 15 KM/Hr    |
| 45 | Turn       | Medium     | 11.46458, 78.47401 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 46 | Turn       | Medium     | 11.46418, 78.47451 | 30 KM/Hr    |
| 47 | Turn       | Medium     | 11.46333, 78.48024 | 30 KM/Hr    |
| 48 | Turn       | Medium     | 11.46211, 78.48159 | 30 KM/Hr    |
| 49 | Turn       | Medium     | 11.46177, 78.48277 | 30 KM/Hr    |
| 50 | Turn       | High       | 11.46097, 78.48362 | 15 KM/Hr    |
| 51 | Blind Spot | Blind Spot | 11.44226, 78.48592 | 10 KM/Hr    |
| 52 | Turn       | Medium     | 11.45575, 78.52476 | 30 KM/Hr    |
| 53 | Turn       | High       | 11.45633, 78.52755 | 15 KM/Hr    |
| 54 | Turn       | High       | 11.45724, 78.52761 | 15 KM/Hr    |

### Emergency Locations

|    | type     | name                                | coordinates            | speed_limit | risk_level |
|----|----------|-------------------------------------|------------------------|-------------|------------|
| 0  | hospital | Tiruchengode, Goverment Hospital    | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1  | hospital | SPM Medical Centre,Tiruchengode     | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5  | hospital | Jayaa Hospital                      | 11.3848223, 77.9021886 | 30 km/h     | Medium     |
| 10 | hospital | Government Hospital, Namagiripettai | 11.4569484, 78.265263  | 30 km/h     | Medium     |
| 11 | hospital | Annai Hospital                      | 11.4437321, 78.4858654 | 30 km/h     | Medium     |
| 12 | clinic   | Shreetejo Ayurveda Clinic           | 11.444915, 78.485868   | 30 km/h     | Medium     |
| 13 | hospital | Vinola Hospital                     | 11.4439899, 78.4857585 | 30 km/h     | Medium     |
| 14 | hospital | The Chirist Hospital                | 11.4443163, 78.4862134 | 30 km/h     | Medium     |
| 15 | clinic   | Sabeetha Surgical Clinic            | 11.4421254, 78.4867012 | 30 km/h     | Medium     |
| 16 | hospital | Dr. Ramu Life Care Hospital         | 11.4426079, 78.4879762 | 30 km/h     | Medium     |
| 17 | hospital | Thai Hospital                       | 11.4424776, 78.4894516 | 30 km/h     | Medium     |
| 18 | clinic   | Jayaraj Memorial Clinic             | 11.4415635, 78.4890421 | 30 km/h     | Medium     |

### Crowded Spots

|   | type        | name                                      | coordinates               | speed_limit | risk_level |
|---|-------------|-------------------------------------------|---------------------------|-------------|------------|
| 2 | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி              | 11.3850439,<br>77.8948279 | 30 km/h     | Medium     |
| 3 | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி             | 11.3844247,<br>77.8949053 | 30 km/h     | Medium     |
| 4 | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை       | 11.3833608,<br>77.8970145 | 30 km/h     | Medium     |
| 6 | marketplace | Farmer Market                             | 11.4146849,<br>78.0805463 | 30 km/h     | Medium     |
| 7 | school      | Government Higher Secondary School        | 11.4150045,<br>78.0838172 | 30 km/h     | Medium     |
| 8 | school      | Government school                         | 11.4248027,<br>78.1323902 | 30 km/h     | Medium     |
| 9 | college     | Paavai Arts And Science College For Women | 11.4262443,<br>78.1546987 | 30 km/h     | Medium     |

## Route Photos of Risky Spots



**Risk Type:** Roundabout

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.38395, 77.89480





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44019, 77.87547



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.39685, 77.88861





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.38346, 77.89975**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.38302, 77.89995**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.38188, 77.90257**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.37830, 77.90160



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.36754, 77.93919





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.41361, 78.08073



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.41453, 78.08062



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.41476, 78.08094





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.42956, 78.11590



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.42450, 78.15278





**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.43542, 78.20754



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45075, 78.20391



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.45099, 78.20397





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44946, 78.22270



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44793, 78.22320





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.44753, 78.22443



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45361, 78.23597



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49628, 78.32342





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.50868, 78.33972



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.51180, 78.34422





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.51040, 78.34875



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.50969, 78.34967



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.50905, 78.35285





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.47977, 78.38044



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.47754, 78.38114





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.46956, 78.39365**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.46783, 78.39421**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.46455, 78.39289**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45614, 78.40096



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45560, 78.40272





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45658, 78.40442



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.45483, 78.40829



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46741, 78.45087





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.46729, 78.45447



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46808, 78.45535





**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.46798, 78.45844



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.46857, 78.45844



**Risk Type:** Turn  
**Risk Level:** High  
**Speed Limit:** 15 KM/Hr  
**Coordinates:** 11.46799, 78.45992





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46458, 78.47401



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46418, 78.47451





**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46333, 78.48024



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46211, 78.48159



**Risk Type:** Turn  
**Risk Level:** Medium  
**Speed Limit:** 30 KM/Hr  
**Coordinates:** 11.46177, 78.48277





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.46097, 78.48362



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.44226, 78.48592





**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45575, 78.52476



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45633, 78.52755



**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45724, 78.52761



